

$$a_0 \sim N(., .); a_1 \sim N(0, .); \dots; a_j \sim N(0, .) \quad \text{sigma} \sim \text{exp}(.)$$

$$\mu = a_0 + a_1 + \dots + a_j$$

$$y \sim N(\mu, \text{sigma})$$

non-adaptive
1-level model

adaptive 2-level
Shrinkage model

meta-prior

$$sd_j \sim \text{normal}(.,.)$$

$$a_0 \sim N(., .); a_1 \sim N(0, sd_1); \dots; a_j \sim N(0, sd_j) \quad \text{sigma} \sim \text{exp}(.)$$

$$\mu = a_0 + a_1 + \dots + a_j$$

$$y \sim N(\mu, \text{sigma})$$

**adaptive
priors**